Predictive spatial codes anchor recurrent direction memory

Completed computational study. All planned cohorts are included: 104 unconstrained RNNs across three separate studies, eight trained designed-ring interface sets, eight original navigation LSTM pairs, and ten published path integrators. Pilot results remain separate.

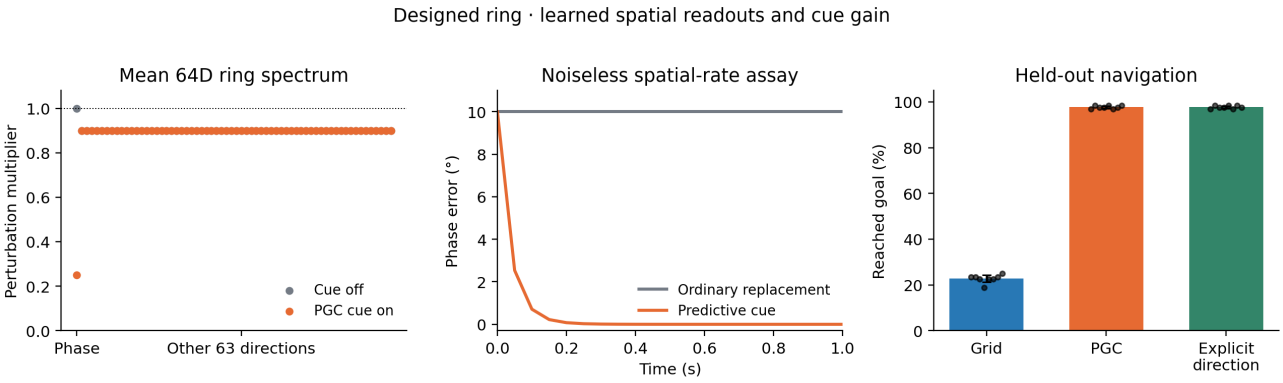
The mechanism is simple: **ordinary grid cells say where the agent is; their relationship to predictive grid cells also says which way it is moving. A trained RNN extracts that direction signal and uses it to correct its internal compass.** Correcting the compass reduces steering error.

We test this in two complementary compass models: an exact ring with a trained sensory interface, and unconstrained RNNs trained to estimate direction. Both drive the same fixed navigation controller. Separate experiments examine the original reinforcement-trained navigation LSTMs and internally predictive neurons in published path integrators. These systems answer different questions and are not pooled.

Detailed methods · Reproduce the study · Mathematical mechanism · Protocols and audits · Related work and biological predictions

1. What we built

First, a transparent ring model. Its 64-dimensional recurrent state contains a continuous circle of valid direction memories. With no cue or angular-motion input, every point on that circle is fixed. Disturbances away from it shrink by a factor of 0.9 per update. We train an interface to read spatial phases from 512 channels generated by the repository's grid-population model, and train the strength with which their relative phase corrects remembered direction. The ring, phase-comparison operation and correction rule are designed; the sensory readout weights and correction strength are learned.



Left: the full-state spectrum has one neutral memory mode and 63 contracting modes. A predictive cue makes the phase mode contract too, while the other modes stay unchanged. Middle: the fitted decoder corrects a displaced memory using actual predictive rates in a noise-free diagnostic. Right: noisy navigation over eight independent trained interfaces; dots are seeds. *Exact model, all interventions and methods.*

In held-out navigation at new locations with a misleading initial heading cue, this model reaches the goal **97.75%** of the time with PGC input and **22.75%** with Grid input. Replacing

the predictive code with ordinary grid values in the frozen PGC model reduces success to **21.68%**. Restoring prediction after two seconds recovers success to **58.98%**. All four prespecified navigation contrasts pass correction across eight seeds. A control with explicit measured direction also reaches **97.75%**, identifying the useful information as direction. With nominal initial heading, the separate descriptive test gives **97.66% for PGC** and **28.52% for Grid**, so the benefit also appears without the misleading initial cue.

The mathematical mechanism is exact in this model: at a cue aligned with the stored direction, the memory-direction eigenvalue changes from `1` to `1 - dt × cue strength × cue magnitude`, while the other 63 eigenvalues remain `0.9`. The trained unit-cue phase multipliers are 0.2 or 0.4. Predictive input therefore corrects errors **along** an otherwise neutral ring. It does not strengthen the already designed contraction **toward** the ring.

Second, unconstrained trained RNNs. These test whether a similar useful input pathway appears when recurrent organization is learned rather than prescribed.

We trained an unconstrained 64-unit RNN to keep track of direction while an agent moves, turns, and temporarily loses its spatial observations. Angular-velocity measurements are noisy and biased, so simply integrating them accumulates error. The RNN receives either ordinary grid codes, a mixture of ordinary and predictive grid codes, or ordinary grids plus an explicit direction cue. The last condition receives the **same noisy direction information used to construct the predictive shift**. The shift points 20 cm along measured travel direction; it does not reveal future turns or plans. In this forward-moving task, travel direction and heading coincide.

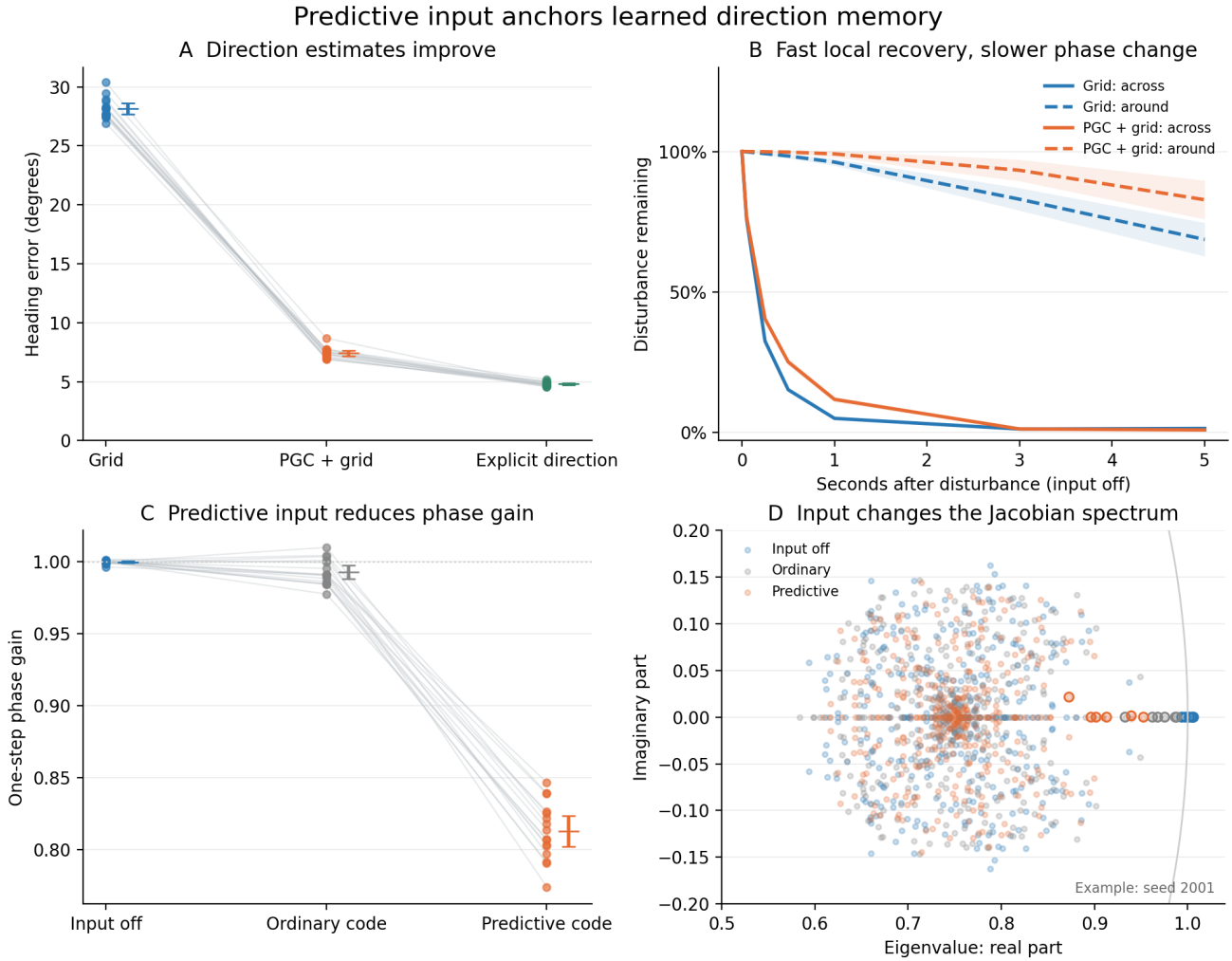
The recurrent weights have no imposed ring wiring. There is no loss that rewards circular hidden states or stable attractors. Training rewards accurate direction estimates. All conditions have the same parameter count, training trajectories, noise, and fixed training budget. The primary study uses **16 paired seeds and 48 trained networks**, with 2,048 new evaluation trajectories per network. An additional eight-seed, 24-network cohort removes the nonlinear sensory encoder.

After training, we examine the entire recurrent state, disturb its memory, change its inputs, and let its direction estimate control navigation. We keep each trained network's weights fixed during these interventions.

2. Predictive input anchors direction memory; full-state geometry is more complicated

Think of the remembered direction as a compass needle. For an ideal ring attractor, moving the state **around** the ring changes the remembered direction; moving it **away from** the ring disrupts the form of the memory. A useful ring should recover from the second disturbance while retaining the first when there is no new information. We test these properties rather than infer them from a circular readout.

We initialize many directions, remove all external input, and perturb the full 64-dimensional state. We measure recovery toward the evolving memory curve and retention of angular displacement. This distinguishes a memory ring from a system that merely collapses every disturbance into the same state. PCA is used only to draw the states; the recovery measurements use all 64 dimensions.

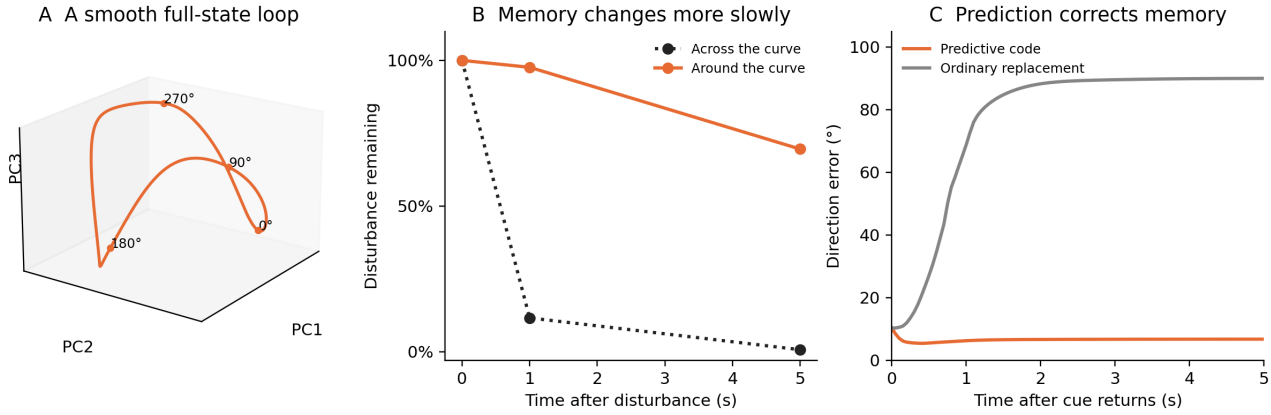


Dots in panels A and C are trained networks; lines and intervals summarize all 16 seeds. “Across” and “around” refer to the sampled memory curve. The spectral panel shows all 64 eigenvalues at six prespecified phases in seed 2001, using identical states under different inputs; larger open circles mark eigenvalues closest to +1. The phase gain is a coordinate derivative, not an eigenvalue.

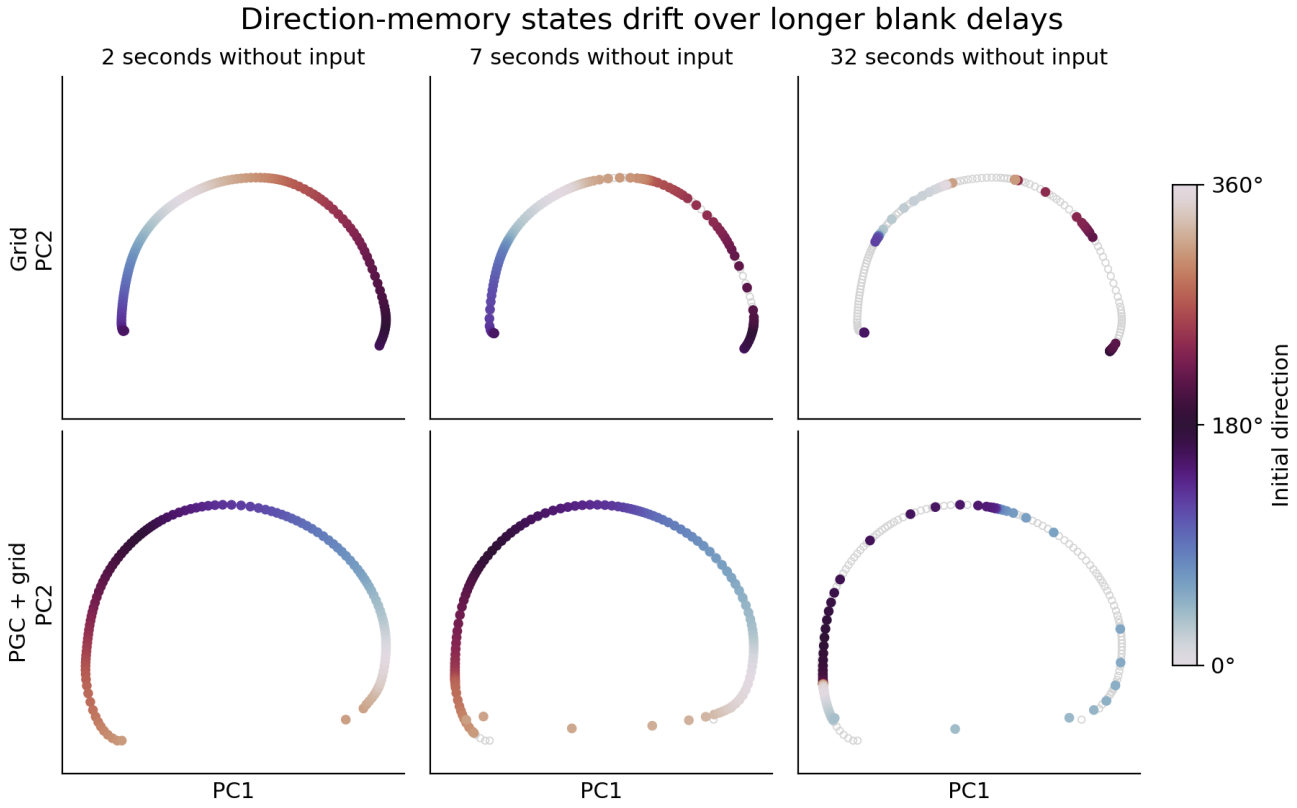
Both populations retain angular memory over short delays and show strong local recovery. Predictive input supplies a reference that corrects drift in the remembered direction. This is the relevant functional stabilization. Long blank delays distinguish it from autonomous persistence: after 32 seconds without input, heading error is **31.62° for Grid** and **44.49° for PGC**. Predictive training does not uniformly improve memory when its directional reference is unavailable.

The full-state audit reveals heterogeneous geometry. Some networks have smooth closed memory loops; others have extremely sharp transitions between distant hidden states, even while decoded angle changes smoothly. Dense sampling resolves these transitions, mostly in directions invisible to the readout. Median eigenvector alignment and sparse circular plots can conceal those exceptions. We therefore test ring-like organization model by model rather than asserting that every network contains a single smooth global ring. Local recovery remains useful evidence, while its precise polyline-based magnitude depends on resolving the sharply transitioning regions.

One learned PGC network, seed 2002, provides a clear example: its full-state curve closes smoothly across 4,096 tested directions, its decoded direction changes monotonically around the loop, and every tested local transverse spectral radius is below one. After one second, mean normal displacement is about 12% of its initial value, while median angular-displacement retention is about 98%. Predictive input then corrects the remembered direction. This is an exploratory example selected after auditing the first seed, not an assertion about every trained model.



The geometry is displayed in three principal components, but closure, derivatives, spectra and recovery use all 64 dimensions. Seed 2002 is the first subsequent candidate examined after seed 2001 revealed sharp transitions. Both examples and every cohort model remain in the analysis.



Prespecified example: first confirmation seed, first position. Color indicates initial direction. The projection is fixed across delays within each model, and gray open circles show its two-second reference states. Points are not connected across unobserved gaps. Dynamical measurements use all 64 dimensions. Dense and bridge-sensitivity diagnostics.

An independent geometry replication limits the broader claim. We trained 32 additional networks and measured how unevenly the full memory state changes as the initial direction rotates. The prespecified peak-to-median sensitivity was lower for PGC in 10 of 16 pairs, but the population difference remained uncertain: mean log10 difference **-0.448**, 95% paired interval **-1.013 to +0.117**, exact **p = 0.110**. All models and the original endpoint were retained. This does not confirm the exploratory suggestion that PGC training generally produces a smoother memory map. The strongest replicated mechanism is the useful correction supplied by predictive input; autonomous geometry varies between networks. Independent geometry results and figure.

At the same recurrent state, changing only the input changes the slopes of the nonlinear activation function. Those slopes change how recurrent feedback amplifies a phase disturbance, even though the recurrent weights are unchanged. The full Jacobian, its eigenvalues and eigenvectors, and actual finite-time recovery provide complementary measurements of that mechanism. An eigenvalue near one alone is insufficient: untrained networks can have one without it aligning with a meaningful direction-memory mode.

All **48 primary networks** completed the fixed training budget and evaluation. Across all 16 paired seeds:

Network	Heading error	Error at new locations
Grid	28.14°	28.13°
PGC + grid	**7.42°**	**7.98°**
Grid + explicit direction	4.80°	4.80°

PGC input reduces error by **20.73°** relative to Grid (95% paired interval 20.26-21.19°). Replacing predictive values with ordinary grid values in the frozen PGC networks raises error to **73.45°**, a cost of **66.03°** (64.58-67.48°). Each of the four locked primary contrasts has Holm-adjusted p = **0.000122**, with the predicted sign in every seed. The additional accuracy-and-coverage anchoring criterion is met by **15/16** models; the remaining model is retained in every analysis. Complete statistics and source data.

The sensory interface matters. The separate 24-network control removes the nonlinear encoder and reduces parameter count, while retaining the same task and training budget. Predictive input still improves tracking: **21.38° for PGC versus 29.08° for Grid**, with **5.28°** for explicit direction. Ordinary replacement raises PGC error to **31.12°**. Tracking, removal and spatial transfer all pass the separate three-test correction across eight seeds (adjusted p = 0.0234). The benefit is smaller than in the larger encoder architecture. Simple architecture-control figure.

Its stationary anchoring test also shows why contraction is not enough: predictive input strongly contracts local phase disturbances, yet its mean stationary heading error is **25.22°** and coverage is **84.3%**. It still improves markedly over ordinary replacement in those same networks. The useful input must be interpreted accurately, not merely drive the state toward something stable. This is an architecture-and-capacity sensitivity, not an isolated test of nonlinearity. Memory and anchoring diagnostics.

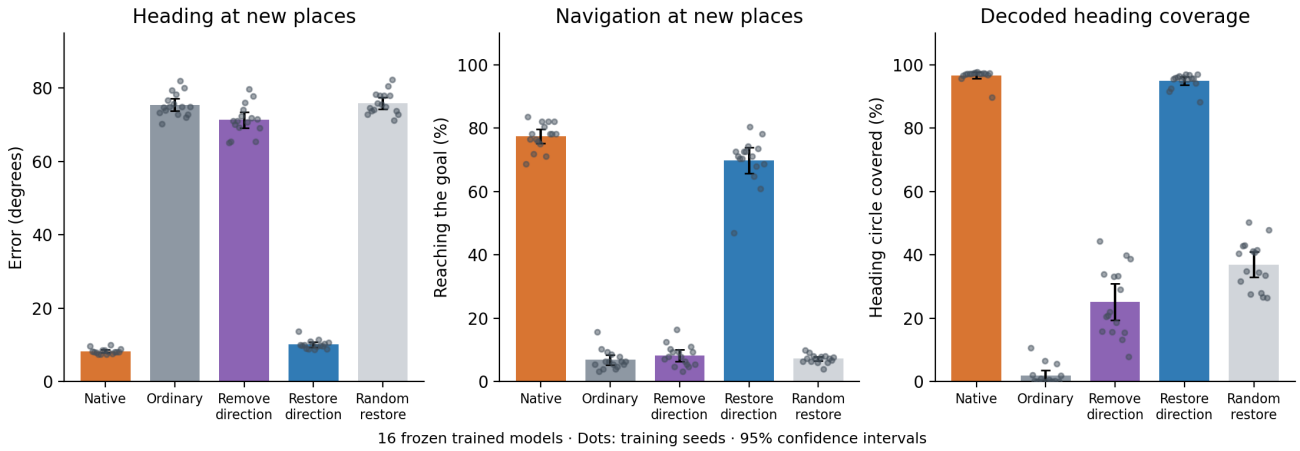
3. Removing and restoring the direction signal

We first replace every predictive input with the ordinary grid value that the same cell would have produced at the same position. This preserves the grid maps, cell identities, input size, and trained weights; it removes the forward spatial shift.

We then make a more selective intervention. Using independent calibration data, we find two directions in the learned sensory drive that track cosine and sine of movement direction. Together these form a **direction-carrying plane**; each direction combines inputs to many recurrent units. We remove just that plane's contribution from the native predictive drive, or restore just that contribution to ordinary drive. The intervention uses the observed input patterns and a fixed projector; it receives no true heading or target answer during evaluation.

Three random planes, each matched in input-drive energy, test whether an arbitrary equally large intervention would work. A separate control restores only the fixed calibration mean of the projected drive difference, or removes its varying component while keeping that mean. These controls distinguish directional information from a simple global offset in the network's input.

The direction plane is tested at new physical locations, shifted 10 m beyond the training region, both during matched trajectories and during navigation. This is transfer across position in the same simulated sensor model; it is not a test of arbitrary new environments or sensory modalities.



All 16 frozen PGC networks, using a new matched test bank distinct from the primary evaluation. The plane is fitted on separate data. Random restoration averages three equally energetic control planes within each model. The right panel measures decoded angular coverage at stationary training-region positions, not full-state ring topology.

At new locations, native input gives **8.2°** heading error and **77.4%** navigation success. Removing the direction plane gives **71.3°** and **8.2%**. Restoring just that plane to ordinary input recovers **10.0°** and **69.8%**; random restoration gives **75.9°** and **7.2%**. All four heading and four navigation contrasts favor the direction component in every seed, with Holm-adjusted p = **0.000122** in each four-test family.

The fixed-mean control resolves a simpler alternative. Restoring only the average projected drive gives **75.2°** error and **6.6%** success. Removing its changing component while keeping that mean gives **71.0°** and **8.5%**. The useful signal lies in the varying direction-carrying component; this fixed average supplies little benefit. Mean-control figure and corrected tests.

4. A small equation predicts the correction

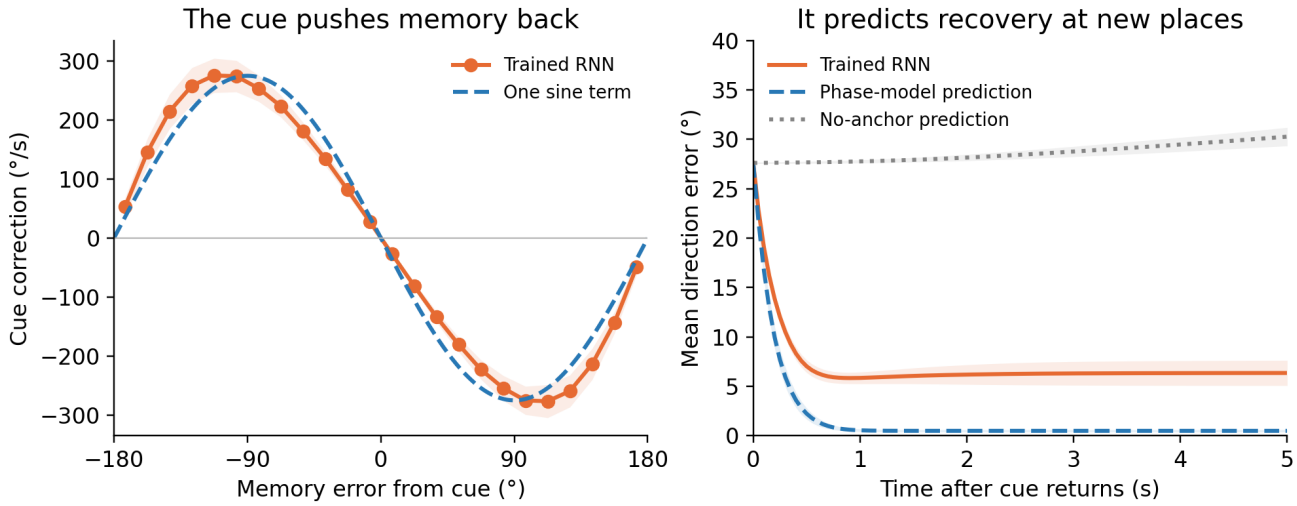
For an ideal grid-wave component, current and predictive spatial phases are `k·x` and `k·(x + d u)`. Their difference is `d k·u`: **position cancels, leaving movement direction**. This calculation explains the information available to the learned encoder; the subspace intervention identifies a functional direction pathway, without uniquely identifying literal phase subtraction as its algorithm. We do not assume that every recorded grid cell has a separately observable sinusoidal phase.

In a stationary-cue assay with no angular-motion or speed input, the recurrent phase dynamics are summarized by

```
rate of change in remembered direction = intrinsic drift + κ × sin(cue direction - remembered direction).
```

We fit intrinsic drift from separate blank-input data and one cue-strength coefficient, κ, from calibration states. We then predict recovery at held-out positions and new starting-state/cue combinations. Reversing the predictive displacement changes the predicted stable direction

without refitting κ. This is a prediction of the network's time evolution, beyond fitting a static activity plot.



The reduced equation is fitted separately from the displayed test trajectories. Predictive and reversed cues are tested with the same fitted cue strength. The ordinary-replacement condition can leave the memory manifold; a phase-only equation is not expected to describe collapse of the full state.

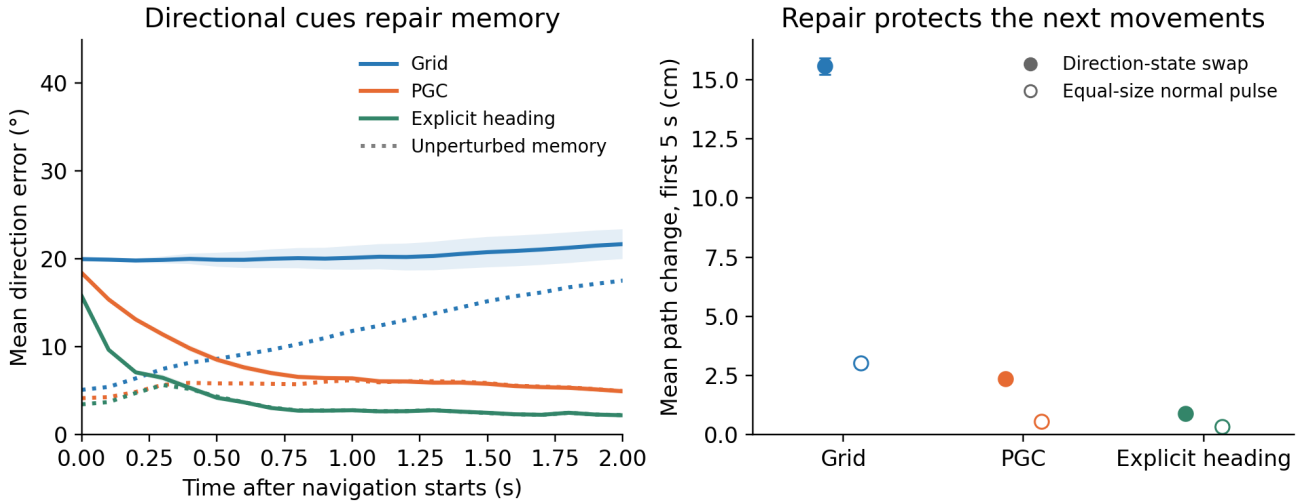
Across all 16 models, the fitted cue strength averages **4.80 per second**. The model explains **78.6%** of held-out phase-field variance. Its recovery-trajectory prediction error is **6.89°**, compared with **66.89°** when the anchoring term is omitted. These are errors in predicting the RNN's dynamics, distinct from its navigation or heading-estimation accuracy.

The reference must carry information that the network does not already have. A predictive shift generated from the network's own mistaken direction estimate can reinforce the mistake. That control separates correction by an independent measurement from a self-consistent but inaccurate internal signal.

5. Does correcting memory help the agent?

Every trained estimator controls the same navigation rule. The controller knows its starting coordinate and goal, then estimates its position from measured speed and the RNN's direction estimate. It never receives current true position or heading. Its own actions determine its subsequent trajectory.

In addition to removing sensory information, we directly substitute a memory state elicited by an initialization cue shifted by 20°. We measure the actual decoded displacement, which need not equal exactly 20°. We compare this with an equally large disturbance away from the sampled memory curve. This tests whether changing the direction-associated recurrent state affects steering and whether predictive input corrects the resulting error. Near a sharp hidden transition, the substitution can also change components invisible to the direction readout; it is not guaranteed to isolate phase alone in every network.



All conditions use the same pulse protocol and controller. Heading recovery and trajectory deviation are separate from eventual arrival. Passing through the goal, remaining there, and stopping there are scored separately.

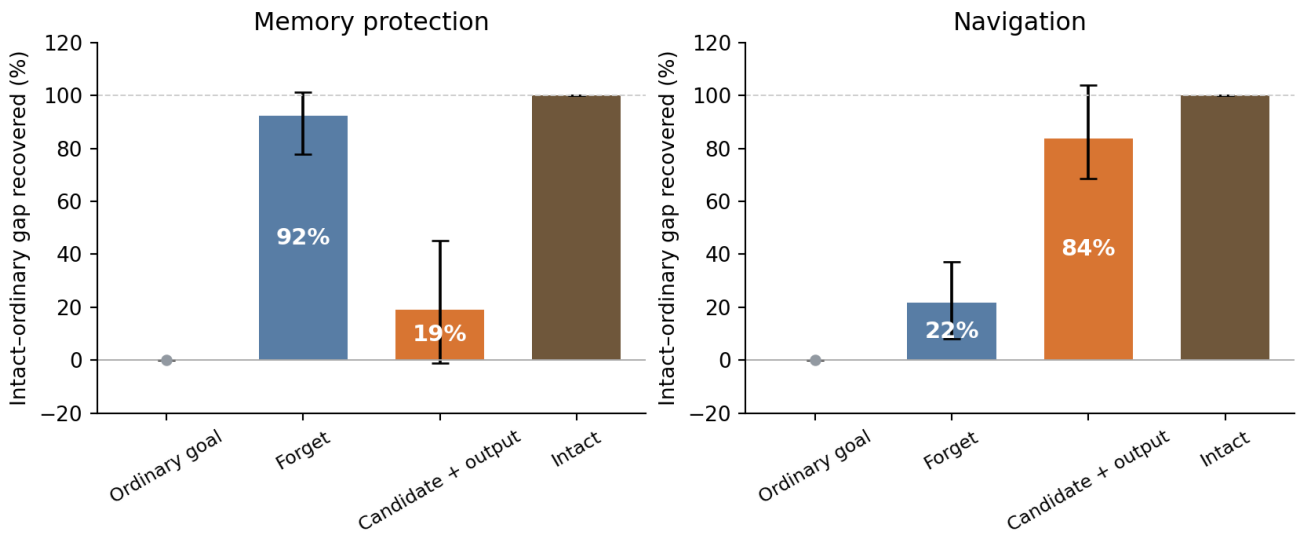
The state replacement changes the agent's path, relative to its own no-replacement trajectory, by an average of **15.58 cm for Grid** and **2.36 cm for PGC** during the first five seconds. The extra heading error caused by the replacement is **4.18° larger for Grid** after two seconds (95% paired interval 3.55-4.81°). The difference in eventual arrival penalties is less consistent: its paired interval includes zero. The clearest behavioral effect is therefore rapid correction of early steering error.

The selective sensory intervention and the direct memory perturbation answer different causal questions. The first locates useful information in the learned input pathway. The second tests the behavioral consequences of displacing the recurrent memory itself.

6. What this explains in the original navigation agents

The original eight pairs of 1B-frame LSTMs show stronger episode-related organization of activity: the first ten PCs capture 46.4% of variance for Grid and 52.3% for PGC+grid. Subtracting each episode's average reduces those values to 40.9% and 42.8%. This describes how activity is distributed across contexts. It does not identify a memory ring, and our autonomous probes did not find the new compass model's ring in those LSTMs.

The original agents nevertheless show a related, specific dynamical effect. Predictive goal input reduces amplification along vulnerable cell-memory directions. Restoring its influence on the **forget gate** recovers about 92% of the measured four-second memory protection. This finer gate partition is exploratory: its exact test has Holm-adjusted $p = 0.094$ across 12 comparisons. Restoring predictive influence on the **candidate update and output gate** recovers about 84% of lost navigation score, supported by the separate corrected behavioral comparison. These percentages come from different assays; they are not additive pieces of a single effect.



Same goal-input gate interventions; separate assays · 8 seeds · 95% paired bootstrap intervals

Predictive current-position input stays intact. Only predictive goal-code features are replaced and selectively restored to specified gate inputs. Eight training seeds; the memory assay uses 32 replay states per model, and the behavioral assay uses 128 matched episodes per condition and model. Full results and exact intervention.

This separates memory protection from the pathway carrying the larger measured navigation benefit. It also explains why “PGC networks are globally more stable” is too broad: average sensitivity, vulnerable memory directions, and useful behavioral computation are different properties.

7. Internal predictive neurons and the broader mechanism

An additional test uses ten published 4,096-unit path-integrating RNNs with predictive-cell identities fixed by their published classifications. We attenuate the recurrent outputs of predictive units and compare them with matched non-PGC controls. We separately replace the removed output with its independently calibrated mean. This asks whether a lesion removes changing information or mainly changes the network's operating point. These internally generated predictive neurons are different from the analytic spatial inputs supplied to the new compass model.

The ten-network replication is complete. At the prespecified 25% attenuation and first 0.4-second endpoint, predictive-cell interventions differ from matched controls by less than 0.1 cm on average in all four conditions; every paired interval includes zero. All five corrected primary tests have $p = 1.0$. The selective deficit seen in the earlier single-checkpoint pilot therefore does not replicate under these published classifications and source-only interventions. Later and stronger interventions show mixed effects. Full results, doses, controls and figures.

A complementary sensorimotor ring model demonstrates another route: forecasting can compensate delays in the agent-environment feedback loop. Its neural ring is designed in advance; its stabilization concerns the joint physical and neural loop. The learned-compass experiment instead identifies an input-driven correction mechanism in an unconstrained trained recurrent network.

The combined account is that predictive codes can improve behavior in more than one way: by providing a directional reference, shaping recurrent sensitivity, and compensating

delayed control. The strongest new causal claim here is the first: **a learned direction-carrying input pathway anchors a recurrent compass and improves navigation.** Its biological prediction is that selectively disrupting the predictive-versus-current spatial relationship should displace or destabilize the downstream direction estimate, even when overall activity is preserved.

Reproduction and scope

The methods appendix specifies source hashes, simulator timing, independent data splits, every intervention, statistical units, and reproduction commands. Primary endpoints were frozen locally before the confirmation cohort; secondary protocols record when they were fixed. Analyses use trained seeds as replicates and retain all planned seeds. Multiple-comparison correction is applied within the declared families, rather than treating every simulated time step as an independent observation.

The new results are computational. The direction reference is deliberately available to the analytic PGC generator; the model explains how it can be used, not how biological PGCs generate it. The explicit-direction control shows the value of the information independently of its grid representation. Prior work already establishes learned compass rings and sensory anchoring; the contribution is the tested chain from predictive spatial code, through a learned sensory subspace and recurrent phase dynamics, to navigation. Literature context.

Methods: predictive input, learned heading memory, and navigation

This appendix details two distinct experiments. The **new heading model** is trained to estimate heading from simulated movement and spatial observations; a fixed controller then uses its estimate to navigate. The **original navigation agents** are previously trained square-arena LSTMs examined with input and gate interventions. Their architectures, tasks, training objectives, and seed cohorts are different and are not pooled. The separate published-network experiment on **internally classified predictive neurons** has its own report and frozen protocol; its ten networks are not part of either cohort below.

Completed cohort: all 16 paired training seeds, 2001-2016, in all three conditions have completed their fixed training and primary evaluations: 48 models. All 16 predictive models have also completed the direction-subspace and fixed-mean follow-ups. Pilot observations remain separate. Completion is recorded in the primary summary, subspace summary, and fixed-mean summary. The original eight-seed gate experiments are complete. The 24-model raw-input and 32-model independent geometry cohorts also completed every planned model and evaluation; the latter's single primary test did not reach significance (exact p = 0.110). Separate later geometry, published-network, and explicitly designed-ring cohorts have their own protocols and status; they are not counted in these 48 models.

1. New heading task and its information sources

The model and simulator are defined in ring_heading_model.py. The spatial observations use the repository's analytic GridPopulation, rather than learned biological-cell recordings. Each cell has a fixed hexagonal map,

$$r_j(x) = \mathrm{clip}\left(\left(\Sigma_k \cos(K_{jk} \cdot (x - \varphi_j)) + 1.5\right) / 4.5, 0, 1\right)^2.$$

There are 128 cells in four equally sized modules. Module spacings are 0.40×1.42^m metres for $m = 0,1,2,3$. Each module's orientation is sampled uniformly over 0-60°, and cell phases are uniform over the corresponding lattice unit cell. Within a paired training seed, the conditions have identical maps and cell identities. The population seed equals the training seed, so populations vary across the 16 seed pairs.

The three training conditions are:

Condition	Spatial observation	Continuing direction information
Grid	All 128 maps evaluated at measured current position	No direct direction cue after initialization
PGC+grid	43 maps evaluated 20 cm ahead along measured travel direction; the other 85 at measured current position	Direction is encoded indirectly in the shifted spatial pattern
Grid+explicit heading	Same ordinary maps as Grid	The same measured cosine/sine direction used to construct the PGC shift, available when the spatial observation is available

PGCs are assigned across modules in counts 11, 11, 11, and 10. Predictive displacement is a fixed **spatial** shift, not a speed-dependent look-ahead time. The task supplies an independent noisy direction measurement to the observation generator. This is the information source that can correct integrated heading; it is not generated from the RNN's own estimate. The explicit-heading condition controls for access to that information in a more directly readable form.

All conditions receive measured angular velocity, measured forward speed, a spatial-availability flag, two initial-heading components, and a heading-availability flag. The complete input has 134 components. During the first four updates, every condition receives the same measured heading cue. After those updates, Grid and PGC+grid lose that explicit cue. When spatial observations are absent, all spatial code components are zero; continuing angular-velocity and speed measurements remain available. In dedicated cue-free dynamical tests, **all 134 input components are zero**, while learned encoder biases remain active.

The agent moves in an unbounded two-dimensional plane with 50 ms updates. Initial positions are uniform in $[-1.25,1.25]^2$ m and initial headings are uniform on the circle. Angular velocity follows a clipped autoregression, $\omega_t = \mathrm{clip}(0.92\,\omega_{t-1} + 0.30\,\varepsilon_t, -2, 2)$ rad/s. Forward speed follows $v_t = \mathrm{clip}(0.98\,v_{t-1} + 0.02\times0.22 + 0.015\,\eta_t, 0.10, 0.35)$ m/s, except at the initial observation and stationary tails. Half the trajectories receive a stationary tail whose start is randomized by the source implementation. Turning is zero when stopped, so retained travel direction equals heading in this task.

At index t, position and heading already include movement from ω_t and v_t. The angular/speed measurement delivered at t refers to that same completed movement. Thus the observation and heading target are temporally aligned. The source is authoritative for exact random-number order and stopping/blackout sampling; the independent navigation tests check timing and information isolation.

Noise or availability term	Setting
Angular-velocity measurement noise	Gaussian SD 0.30 rad/s per update
Angular-velocity sensor bias	Gaussian SD 0.10 rad/s, fixed within a trajectory
Forward-speed measurement noise	Gaussian SD 0.01 m/s; measured speed clipped below at zero

Noise or availability term	Setting
Direction measurement noise	Gaussian angular SD 0.05 rad
Position measurement noise	Gaussian SD 0.01 m per coordinate
Rate-channel measurement noise	Gaussian SD 0.03, followed by clipping to [0,1]
Spatial availability	Independent probability 0.50, plus a continuous randomized blackout
Training recurrent preactivation noise	Gaussian SD 0.01 per unit per update

The continuous training blackout begins between one-quarter and one-half of the sequence and lasts between approximately one-eighth and three-eighths of it. Some stopped tails remain cued and others become blank. The first four observations are always visible. The same trajectory, sensor noise, channel noise, visibility, and recurrent-noise realizations are paired across conditions within each training seed.

2. Architecture, objective, and fixed training budget

The input encoder is `134 → 256 → 256 → 64`, with SiLU between its linear layers. The recurrent state has 64 unconstrained units:

```
h_(t+1) = 0.75 h_t + 0.25 tanh(b(x_t) + W_rec h_t + ξ_t).
```

An affine two-output readout is trained against cosine and sine of the current true heading. Every condition has **121,026 trainable parameters**. Recurrent weights begin with an orthogonal initialization; encoder weights use Xavier initialization. The architecture has no circular connectivity constraint, state-geometry loss, ring target, or attractor regularizer. Circular heading is the supervised variable, so successful training may organize memory around that variable; hidden-state ring geometry is measured afterward.

train_ring_heading.py minimizes mean squared readout error plus $10^{-4} \times \text{mean}(h^2)$. AdamW uses learning rate 10^{-3}, weight decay 10^{-5}, global gradient clipping at 1, and a cosine schedule ending at learning rate 2×10^{-4}. Each model receives exactly 6,000 updates, batch size 64, sequence length 128. Initial hidden state is zero. The condition-independent task generator uses seed `training_seed + 1,000,003`; model initialization uses the training seed. The intermediate 3,000-update and final 6,000-update checkpoints are fixed in advance. Validation does not select the final checkpoint.

The confirmation protocol was locally timestamped before training seeds 2001-2016. It is not an external preregistration. The source manifest records the protocol and training-source hashes. Checkpoints record the complete configuration, source hashes, seed, optimizer, training history, completed updates, and stopping reason. All planned seeds remain in the analysis, including unsuccessful training; interrupted jobs must complete their original budgets. Earlier raw-input and other exploratory models are separate sensitivity cohorts.

3. Heading evaluation and statistical units

The locked primary evaluator uses four fresh evaluation seeds, **71001-71004**, with 512 trajectories per seed: **2,048 trajectories per trained network**, each 256 steps long. Evaluation includes sensor noise but no added recurrent noise. Native and ordinary-replacement inputs use exactly the same underlying physical trajectories and measurement noise. Primary errors exclude exactly the first **four** initialization updates, unlike some pilot and subspace assays that exclude eight.

Heading error is the absolute wrapped angular difference between the affine readout's `atan2` angle and the true heading. Errors are averaged over eligible times within each trajectory, then over trajectories within each trained seed. Translation tests add `[10,0]` m to both true and measured positions while retaining movement, direction, noise, and timing. An explicit bound verifies that translated true positions lie beyond the training task's reachable x range.

The four primary paired endpoints are PGC-versus-Grid tracking error, ordinary-replacement-versus-native error within frozen PGC networks, native-versus-ordinary phase anchoring, and PGC-versus-Grid translated-position error. The aggregator requires all 16 planned seeds and validates protocol, source, and checkpoint identity. It reports mean paired differences, nominal 95% paired-t intervals, exact two-sided sign-flip tests over all `2^16` assignments, and Holm correction across these four endpoints. Training seeds are the independent units; trajectories, contexts, and phase samples are not additional training replicates.

The transfer evaluator supplies separately labeled secondary reversal, direction-shuffling, restoration, and doubled-noise tests. Its defaults are seed 931731, 256 trajectories × 256 steps, exclusion of eight initial steps, and restoration at step 64. These results do not replace the primary endpoints.

4. What qualifies as a memory ring or as phase anchoring

A circular plot or a low-dimensional PCA projection is not the operational criterion. evaluate_heading_memory.py initializes 128 evenly spaced headings at three physical positions, supplies the four-step training cue, then applies 40 all-zero-input settling steps followed by 600 further blank steps. This tests 30 seconds of continuation after settling. It evaluates the trained model and an architecture-matched untrained model initialized with the same seed.

The analysis uses the full 64-dimensional recurrent state. It measures state radius, decoded-heading coverage and accuracy, winding, readout amplitude, and the distribution of local phase gains. Tangents are cyclic central differences along the measured state curve. Random normal perturbations (seed 419) are projected into the tangent's orthogonal complement. Tangent and normal disturbances have equal norms of 2%, 10%, or 30% of the initial full-state curve radius. Perturbed paths are compared with the **time-matched** unperturbed curve, using nearest projection onto its closed piecewise-linear interpolation as well as direct paired-state distance.

A finite-time ring interpretation requires a persisting, noncollapsed family of states, relatively preserved displacement along that family, and recovery toward it after transverse displacement. Report the entire trajectory and phase distribution; a circle can deform or

gradually drift toward a few preferred states. The geometry routines include a known ring as a positive control and a collapsing input-driven circle as a negative control. Both contract disturbances, but only the first preserves a finite memory manifold.

A subsequent descriptive curve-bridge audit checks an additional limitation: uniform sampling of initialization headings need not sample full-state arc length uniformly. A few adjacent states can differ sharply in hidden dimensions while their decoded directions remain close. Near such transitions, a coarse central difference is a secant rather than a validated local manifold tangent, and a straight interpolated segment may cross unoccupied states. The audit records maximum adjacent jump relative to full-state radius, arc length concentrated in segments longer than five times the median, slow-mode alignment near those segments, and recovery-distance sensitivity after removing only their interpolated interiors while retaining every measured vertex. This threshold is an exploratory numerical diagnostic, not a new confirmatory classification rule. Local recovery and useful direction correction can remain strong even when an unqualified smooth global-ring interpretation fails. Dense-angle and common-history checks are therefore considered separately from angular readout coverage. Audit data.

The separate anchoring assay uses 72 initial headings at five in-region positions and `[10,10]` m. After four cue steps and 40 blank steps, a ±10° phase disturbance substitutes actual neighboring full states from that measured curve. Baseline and disturbed replicas then receive the same physical input. Conditions include native, all-blank, ordinary replacement, reversed predictive shift, and ordinary input followed by restoration after 20 updates. For PGC networks, a self-generated-preview branch constructs each replica's predictive direction from its own previous readout, removing the independent external direction reference.

The locked endpoint is the median per-state retained phase displacement at update 100, averaged across the five in-region positions. The complete continuation is 200 updates. Accuracy, full-state error, state radius, and coverage are reported alongside correction. Accurate anchoring additionally requires native coverage at least 0.90 and lower true-heading error than ordinary replacement. Failure of that criterion remains an outcome, not an exclusion. Convergence into one wrong state counts as collapse. A fixed directional cue pins one phase of the memory ring; it is not expected to preserve a continuum of neutral phases while that same cue remains fixed.

5. Encoder direction-plane intervention

The subspace follow-up protocol and its frozen source manifest were fixed after the pilot and before confirmation subspace outcomes. All 16 PGC models are included, regardless of their geometry or performance. Validation-only and local-phase-summary amendments are recorded in that manifest.

heading_direction_subspace.py computes the frozen encoder-drive difference `Δb = b(x_PGC) - b(x_ordinary)` for paired inputs generated at the same measured position and direction. For fitting, speed/angular velocity are zero, spatial availability is one, and the initial-heading cue is absent. Calibration uses seed **517913**, 1,024 independent positions in $[-1.25,1.25]^2$, and 32 uniformly spaced directions with independent position-specific phase offsets. A regression on an intercept, cosine, and sine supplies two coefficient vectors; their orthonormal span defines the fixed rank-2 projector `P`. The intercept is excluded from this plane.

Held-out sets contain 256 new positions × 32 directions, both in-region and shifted 10 m along x. Report harmonic regression variance explained, within-position directional variance explained, projected drive energy, and held-out circular direction readout error. Rank two is imposed by the first-harmonic question; the complete encoder difference is not assumed intrinsically two-dimensional. Harmonics 1-3 are a predeclared fit-only secondary diagnostic.

Frozen recurrent networks receive native drive, ordinary drive, `b_PGC - PΔb`, or `b_ordinary + PΔb`. The residual-only expression `b_ordinary + (I-P)Δb` provides an algebraic check on the deletion branch. Three rank-2 random planes, generated with seed **517914** and orthogonal to the fitted plane, provide deletion and restoration controls. Their components are scaled to match the fitted component's norm **for each input in encoder preactivation space**. This does not guarantee matched post-tanh state change or readout sensitivity.

Only calibration and scoring use direction labels. The intervention receives two observed encoder drives and a frozen projector, not true heading, position, or a target answer. The experiment supplies the counterfactual ordinary input; it does not assume the unmodified agent naturally receives both vectors as separate channels.

The plane is tested in the 72-phase anchoring assay, 256 matched 256-step forced trajectories with evaluation seed **931731**, and closed-loop navigation. Forced-trajectory comparisons share four native initialization steps and exclude the first eight steps from error summaries. Subspace navigation uses all 16 models, 128 episodes each, seed **836219**, 40 seconds, and native-region and 10 m transfer scenarios. Each navigation branch controls its own physical trajectory. The primary subspace heading and navigation outcomes are at transferred places. Their four paired contrasts-native benefit, deletion cost, restoration benefit, and restoration over the mean random control-have separate prespecified four-test Holm families. The runner and aggregator wait for every seed before producing complete-cohort tests.

Validation checks untouched native/ordinary full-state trajectories, deletion/residual equality, projector idempotence, matched sham norms, zero intervention during blackouts, and unchanged checkpoint hashes. Projector archives are bound to their originating checkpoint SHA and model seed. Independent navigation reference runs require identical success and arrival, with continuous-distance tolerance 10^{-4} m. Exact sources and episode-level data are archived.

A validation-only numerical amendment was needed for the final model, seed 2012. Different float32 arithmetic order in the algebraically equivalent deletion and residual-only expressions accumulated 1.79×10^{-5} maximum hidden-coordinate discrepancy over 256 updates. Float64 reduced it below 6.2×10^{-15}; the float32 decoded discrepancy was below 0.002°. A new documented wrapper changes only this unused alias's assertion tolerance from 10^{-5} to 10^{-4}. Original evaluator bytes, scientific computations, primary branches, other checks, statistics, and all 16 included seeds remain unchanged. The original failing log, effective-function hashes, and independent numerical audit are retained. Shared native, ordinary, and full-plane restoration outcomes match exactly between the two complete subspace studies.

An additional fixed-mean diagnostic, specified after the pilot and after six confirmation models had their original subspace analyses computed, separates a tonic offset from the varying plane component. It uses the original independent calibration mean `μ=mean(Δb)` and

the observed availability flag a. The tonic component is aPμ; the varying component is PΔb-aPμ. New branches restore only the tonic component to ordinary input or remove only the varying component from native input. They preserve exact no-op behavior during blackouts and receive no direction label at inference. All 16 PGC models enter this additional diagnostic, with the same trial seeds, sample sizes, and assays. Three exploratory contrasts per endpoint have their own correction and leave the earlier four-plus-four families unchanged. This control addresses a global fixed offset, not every context-dependent average or nonlinear response to drive fluctuations. Diagnostic implementation · Runner.

6. Jacobian eigenvalues, eigenvectors, and phase gain

For fixed input, the full-state update Jacobian is

$J(x,h) = 0.75\, I + 0.25\, \mathrm{diag}(1 - \tanh^2[b(x)+W_{rec}\, h])\, W_{rec}$.

The cue-free memory analysis samples spectra at absolute blank updates 40, 100, 140, 300, and 340. It reports the eigenvalue closest to +1 separately from the largest-modulus eigenvalue, their alignment with the empirical tangent, full singular values, and projected normal quantities. For a complex eigenmode, tangent alignment uses its real invariant plane rather than a phase-dependent complex eigenvector angle. A normal-to-normal projected singular value excludes leakage into the tangent and must not be interpreted as total growth of all transverse disturbances.

evaluate_heading_cue_spectra.py compares blank, native, and ordinary inputs at **identical** cue-free states and, separately, identical native-anchored states. It uses 72 phases and the six anchoring positions. Native-anchored states follow 200 native updates from the cue-free curve. Changing input while holding state and recurrent weights fixed isolates the change in the activation slopes that multiply recurrent feedback.

For affine readout $y = Wh+b$, the phase gradient is $(y_0 W_1 - y_1 W_0)/||y||^2$. One-step phase gain is

$[\mathrm{grad\ phase}(h_{next}) \cdot J(x,h) \cdot \mathrm{tangent}] / [\mathrm{grad\ phase}(h) \cdot \mathrm{tangent}]$.

This is gain in the trained phase coordinate, not an eigenvalue. The denominator accounts for local conversion from state displacement to phase displacement. Report resolvability and output amplitude, and distinguish instantaneous operating-point spectra from finite-time recovery. The independent audit checks the derivative against autograd and central differences, invariance to tangent scale/sign and readout rotation/scaling, and a system with known radial and phase contraction. It also checks the full-state Jacobian identity through the tanh slopes.

7. Navigation controller and behavioral outcomes

Navigation is a downstream test of the trained heading estimate, not a further reinforcement-learning phase. evaluate_ring_heading_navigation.py supplies the same fixed controller to every model. The controller knows the reset coordinate and goal. It integrates measured forward speed along decoded heading to maintain its own position estimate; it never receives current true position or true heading. The simulator uses true pose only for physical evolution, sensory generation, and scoring. The subspace navigation wrapper retains that controller and simulator.

Starts are uniform in $[-0.5,0.5]^2$ m. Goals are 0.8-1.5 m away at a uniform random bearing. Goal-bearing error is computed from estimated position and decoded heading. Angular command is $\mathrm{clip}(2 \times \mathrm{error}, \pm 1.8)$ rad/s. Forward command is $0.25 \times \mathrm{clip}(\cos(\mathrm{error}), 0.24, 1)$ m/s. The controller stops within 0.08 m of its **estimated** goal. Physical angular velocity and speed use smoothing coefficients 0.8/0.2 and motor-noise SDs 0.03 rad/s and 0.005 m/s per update. Moving speed is clipped to 0.001-0.35 m/s; both speed and rotation are zeroed when stopped. No ground-truth arrival signal resets or stops the controller.

Navigation supplies the standard noisy observations with spatial visibility probability 0.5 and a shared blackout from 4-7 s. Four native initialization steps precede the intervention. The current observation describes the preceding completed movement; the position estimate is updated before choosing the next action, with the initial no-movement observation handled separately. Conditions share underlying per-step observation and motor noise, but their actions produce different physical paths.

First-arrival success means entering a 0.15 m goal radius by 20 or 40 s. The 20 s outcome is a prefix of the same 40 s episode. Also report capped arrival time, final goal distance, traveled distance, and position-estimate error. The secondary navigation protocol distinguishes passing through from holding at the goal: maximum post-entry excursion among entrants, occupancy during the final five seconds across all episodes, and two consecutive seconds inside the goal with speed at or below 0.05 m/s. These are score-only measurements. Scenarios additionally include doubled angular bias, a ±20° misleading initial heading cue, and their combinations with spatial translation. Subspace confirmation fixes its scope to baseline and translated places.

8. Independent geometry replication and architecture sensitivity

The independent geometry protocol was fixed after exploring the primary cohort's geometry and before training new seeds 5001-5016. It trains 16 new Grid/PGC pairs with the unchanged encoder, task and 6,000-update budget. At position (0,0), each model receives four deterministic initialization cues followed by 40 all-zero-input steps. The analysis propagates exact derivatives of the full state with respect to initial heading in float64, using 1,024 uniform angles and 20 adaptive bisections in each of four large-separation intervals. The primary endpoint is log10 of the estimated peak derivative norm divided by its uniform-angle median. All 32 models are retained; fixed floors and degeneration flags are recorded.

This measures unevenness of the initialization-to-memory map. It is not a count of attractors, a topological classification, or an assay of transverse contraction. Absolute peak and median sensitivities, state radius, readout coverage and accuracy accompany the ratio. Cue-end and matched ordinary-initialization conditions are descriptive controls. The single paired contrast uses all 16 seeds, a nominal 95% paired-t interval, and an exact two-sided sign-flip test over all 65,536 sign patterns. Aggregation waits for all pairs. The frozen evaluator checks independently propagated derivatives against automatic differentiation in each model, and the final-cohort audit reconstructs the endpoint from saved arrays.

The separate raw-input protocol uses eight new seeds, 3001-3008, with all three input conditions. It replaces the nonlinear encoder with a linear sensory map while retaining 64 recurrent units, the same task, and the same 6,000-update budget. This reduces parameter

count: it is an architecture and capacity sensitivity, not an isolated manipulation of nonlinearity. Parameter count remains matched across input conditions within the cohort. Tracking, predictive removal, and spatial transfer use the unchanged 2,048-trajectory evaluation and a separate three-test Holm family over eight paired seeds. Raw-input memory and anchoring diagnostics are descriptive. These 24 models are never pooled with the primary 48 or geometry 32.

The explicitly designed ring uses another eight-seed cohort, 4001-4008, with its own fixed protocol and report. Its sensory phase readouts and scalar cue gain are trained; its ring geometry and correction law are imposed. Its exact mathematical derivation appears after these methods in the PDF and is also available separately.

9. Separate original-agent LSTM gate experiment

The original trained navigation LSTMs use seeds **1-8**, distinct from the new heading cohort. Their completed input and gate experiments are documented in predictive-memory-gates.md, with source/data links and all statistical families.

The targeted stability assay takes 32 native replay states per PGC model, computes the leading right singular vector of the full `(h,c)` update under native or ordinary-goal input, and tests each fixed direction under both inputs. Perturbations have raw-coordinate RMS amplitudes 10^{-4} and 10^{-2}, equivalent to Euclidean norms of those values times $\sqrt{512}$ for the concatenated 256-unit hidden and 256-unit cell states. Replicas follow 60 actual subsequent input updates and are compared with their own condition-matched unperturbed branch. One-update and 60-update amplification form four planned paired contrasts. These selected vulnerable directions complement a separate random-perturbation experiment; they are not a claim about average noise sensitivity.

Gate attribution restores selected **input preactivation rows** while each branch retains its own evolving state and unchanged recurrent weights. It decomposes the cell update into direct carried-state, forget-feedback, input-gate, and candidate terms. Twelve exploratory gate comparisons are corrected separately. This analysis identifies memory sensitivity pathways; it does not by itself identify a ring.

Matched behavioral tests use four conditions: intact predictive goal input; ordinary goal replacement; ordinary goal with forget-gate restoration; and ordinary goal with candidate-plus-output restoration. Current predictive spatial input remains intact throughout. Only the 169 predictive goal slots are converted to ordinary code. There is no added state noise in this behavioral assay. Evaluation seed **737373**, eight models, four conditions, and 128 matched episodes per model-condition produce **4,096 completed episodes**. Layouts, physical noise, and action random uniforms are shared; trajectories and recurrent states evolve separately. Four planned paired score contrasts use the eight training seeds as units, with separate Holm correction of paired-t and exact sign-flip tests. Memory-protection and navigation-recovery percentages refer to different assays and are not additive gate contributions. A behavioral disturbance test would be needed to link the measured protection directly to navigation benefit.

10. Reproduction and provenance

Training, evaluation, and aggregation entry points are linked above. A single example model can be reproduced with:

```
python scripts/train_ring_heading.py --condition predictive --seed 2001 \
--encoder-size 256 --hidden-size 64 --steps 6000 --length 128 \
--batch-size 64 --learning-rate 0.001 --study-stage confirmatory \
--device cuda --out reproduction/predictive_s2001.pt
python scripts/evaluate_heading_confirmation.py reproduction/predictive_s2001.pt \
--out reproduction/primary_predictive_s2001
python scripts/evaluate_heading_anchor.py --checkpoint reproduction/predictive_s2001.pt \
--out reproduction/anchor_predictive_s2001
python scripts/evaluate_heading_memory.py reproduction/predictive_s2001.pt \
--out reproduction/memory_predictive_s2001
python scripts/heading_direction_subspace.py --checkpoint reproduction/predictive_s2001.pt \
--out reproduction/subspace_predictive_s2001
python scripts/heading_subspace_navigation.py --checkpoint reproduction/predictive_s2001.pt \
--subspaces reproduction/subspace_predictive_s2001/fitted_subspaces.npz \
--out reproduction/subspace_predictive_s2001/navigation --validate-reference
```

Use all three training conditions and all seeds 2001-2016 for the complete paired study. CPU is also supported. Training refuses to overwrite an existing checkpoint/log. The locked evaluator rejects mismatched source hashes, configurations, incomplete training, and pilot identities. Checkpoint bytes and exact source snapshots, rather than a claimed hardware-independent bitwise retraining guarantee, identify the analyzed models. Frozen protocols, intermediate checkpoints, numerical controls, per-episode data, and per-seed summaries make the reported comparisons traceable. The reproduction guide documents the public final-checkpoint archive, dependencies, tests, and regenerated figures.

Why the predictive cue stabilizes ring phase

This derivation describes the **explicitly designed ring model**, not an inferred exact equation for the unconstrained trained RNNs. Its sensory readouts and cue gain are learned; its continuous-attractor geometry and correction law are imposed. The implementation, eight-seed experiment, and exact-map checks accompany the derivation.

1. The spatial code contains a direction difference

For one ideal Fourier component with wave vector k, ordinary and predictive spatial phases are

$$\psi_{current} = k \cdot x,$$

$$\psi_{predictive} = k \cdot (x + d\,u),$$

where $u = (\cos\theta, \sin\theta)$ is measured travel direction and $d = 0.20\ m$. Consequently,

$$\psi_{predictive} - \psi_{current} = d\,k \cdot u.$$

The common position cancels. Stacking non-collinear components gives $\Delta\psi = G\,u$, with rows of G equal to $d\,k^{T}$; a known-geometry pseudoinverse recovers the two-component direction vector. Using a wrapped phase requires $|d\,k \cdot u| < \pi$ for the selected components. The constructed decoder therefore uses the coarser modules 1-3. The finest module can violate this condition and create a direction alias.

The actual rate channels are the repository's squared three-cosine grid functions. We fit linear readouts of the complex phase components from independent calibration examples; no phase is directly supplied at inference. The fitted current and predictive complex readouts are multiplied as $z_{predictive} \times \mathrm{conjugate}(z_{current})$, and their arguments provide the relative phases. The original position or direction labels do not enter the inference call.

This cancellation explains an available computation. It does not identify literal phase subtraction as the unique algorithm learned by the separate unconstrained encoder. Nor does it imply that ordinary grids lack direction information in their temporal history. The constructed Grid control uses the same instantaneous phase-readout architecture.

2. A genuine ring in a 64-dimensional state

Let U be a 64×2 matrix with orthonormal columns. Write any state with a nonzero phase-plane component as

$h = U\,q + v$, with $U^{T}v = 0$, $q = r\,u(\varphi)$, and $r > 0$.

There are two coordinates in q and 62 in v. The ring is the set

$$M = \{\, U\,u(\varphi) : \varphi \in [0, 2\pi) \,\}.$$

The exact update first predicts the completed angular movement,

$$\varphi_{prior} = \varphi + \Delta t\, \omega_{measured},$$

then assimilates the current decoded sensory vector z:

$$\varphi_{next} = \varphi_{prior} + \Delta t\, \kappa \det(u(\varphi_{prior}), z),$$

$$r_{next} = 1 + \rho(r - 1),$$

$$v_{next} = \rho v,$$

$$h_{next} = U\, r_{next}\, u(\varphi_{next}) + v_{next}.$$

Here $\rho = 0.9$, $\Delta t = 0.05$ s, and learned κ has units per second. The decoded cue vector is bounded in magnitude at 1.5 and retains its original magnitude near zero. An ordinary-code difference close to zero therefore remains a weak signal. The cue is zero when unavailable. Recurrent noise is added separately in the noisy navigation experiment.

With both cue and angular-motion input absent, $\varphi_{next} = \varphi$. Every point on M is fixed, while radial error and all 62 orthogonal coordinates shrink by ρ. The ring is therefore transversely attracting and neutral along its phase, on the domain $r > 0$.

3. The cue changes exactly one local stability direction

Consider a stationary cue $z = m\, u(\theta)$ and a state on the ring aligned with it. For phase error $e = \varphi - \theta$, the exact angular map is

$e_{next} = e - \alpha \sin e$, where $\alpha = \Delta t\, \kappa\, m$.

Linearizing at $e = 0$ gives the phase multiplier $\lambda_{phase} = 1 - \alpha$. The radial multiplier and the 62 other multipliers remain ρ.

For the unit full-state tangent $t = U\, u_{perp}(\theta)$, the aligned Jacobian is

$$J_{cue} = \rho I + (1 - \alpha - \rho)\, t\, t^{\top}.$$

Without the cue,

$$J_{blank} = \rho I + (1 - \rho)\, t\, t^{\top}.$$

Thus the cue changes the Jacobian by the rank-one term

$$J_{cue} - J_{blank} = -\alpha\, t\, t^{\top}.$$

This is the mechanism in full neural coordinates: **the cue supplies restorative feedback along the memory direction while leaving transverse recovery unchanged**. The eight trained interfaces select $\kappa = 12$ or 16, so an aligned unit cue gives phase multipliers 0.4 or 0.2. The blank spectrum is $\{1, 0.9 \times 63\}$.

Alignment matters. At a phase error e, the angular-coordinate derivative is $1 - \alpha \cos e$; the full-state Jacobian away from a fixed point is not obtained by simply reusing the aligned eigendecomposition. A reversed cue makes the direction opposite the true heading stable. Contracting errors toward that wrong cue is stable but inaccurate memory. This reversal uses the frozen trained decoder: a decoder trained from the outset on a backward-shifted

code could learn its opposite sign. The result demonstrates dependence on the trained cue relationship, not an exclusive advantage of forward over backward codes.

For a correct constant cue and $0 < \alpha < 2$, every angular error strictly between $-\pi$ and π, except zero, decreases in absolute value under the deterministic map: $|e - \alpha \sin e| < |e|$. The exactly opposite state is an unstable exception. This elementary convergence result applies to the designed map; it is not a theorem about the unconstrained 64-unit RNNs.

4. Why an independent cue controls accumulated error

For small errors, a unit directional cue and angular measurement noise give the approximate update

$$e_{next} \approx (1 - a)(e + \Delta t\, b + \Delta t\, \varepsilon_\omega) + a\, \varepsilon_\theta + \varepsilon_{recurrent},$$

where $a = \alpha$ on visible-cue steps and zero otherwise, b is angular-sensor bias, and the three ε terms represent independent errors. If availability is independent with probability p, the mean error multiplier is $1 - p\alpha$. With a constant bias, the corresponding steady mean is

$$\mathrm{mean}(e) \approx [(1 - p\alpha)/(p\alpha)]\, \Delta t\, b.$$

Without a reference ($p\alpha = 0$), bias accumulates instead of approaching this finite mean. In the zero-bias, small-error case, mean-square stability requires $E[(1 - a)^2] < 1$, giving the same local bound $0 < \alpha < 2$ when $p > 0$.

This approximation explains the tradeoff controlled by the learned gain: stronger correction reduces integrated drift but passes more cue noise into memory. It is an analytical interpretation, not a fitted quantitative prediction of every noisy navigation trajectory. A cue generated solely from the network's own mistaken estimate does not provide the independent information assumed here.

5. Numerical precision and connection to the learned RNNs

The spectrum audit uses an explicitly re-orthonormalized double-precision basis and matches the analytic spectrum to approximately 2×10^{-15}. It also uses cue magnitudes decoded from actual spatial-rate inputs. The operational float32 model is checked separately: its noise-free circle drifts by less than 8×10^{-7} in full-state distance over ten seconds. This separates the mathematical design from finite-precision arithmetic.

The unconstrained learned RNNs have no imposed U, radial law or rank-one Jacobian change. Their measured Jacobians change through input-dependent activation slopes, and their geometry can include sharp hidden-state transitions. A learned direction-carrying input plane, same-state Jacobian comparisons, held-out phase-field predictions and behavioral interventions provide empirical links to the simple mechanism. The designed model supplies a controlled existence proof; it does not replace those empirical checks or turn every learned network into the same ring.